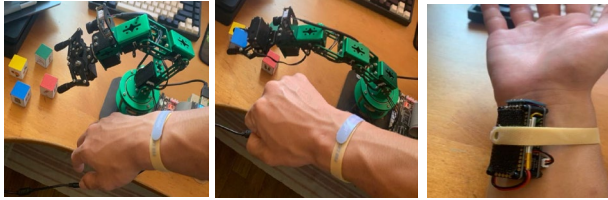


Q1: Controlling 6DOF Robotic ARM via IMU Sensors

An IMU sensor board is used to control a robotic arm such that the robotic arm will emulate the movement of the human controller. IMU data are also used to train an NN model for gesture recognition.



IMU data sampling rate used is 100Hz. Initial testing shows the following accelerometer and gyro data excitation when performing the required gestures (Figure Q1-1). Assume the two graphs are representative of all gestures required.

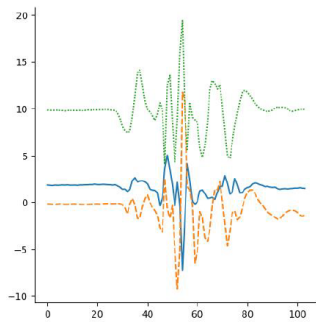


Figure 9 Accelerometer Waveform

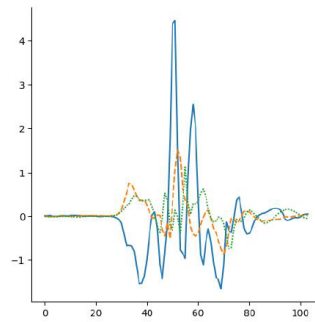


Figure 10 Gyroscope Waveform

Figure Q1-1: Accelerometer and Gyro Data

The signal waveform occupies about 100 samples, 1 second, thus good capture window is about 1 second, window size for training can be smaller at about 0.6-0.8, the capture size can be different if we use sliding window technique

(i) What is a good window size to use for data capture?

(ii) Does the dataset require any pre-processing to enable a more effective recognition model?

Data normalisation

(iii) Data augmentation is often used to increase the size and randomness of the training dataset. For this system, in addition to adding random noise, one way is to offset the main pulse (excitation) in Figure Q1-1 horizontally. In doing so, part of the pulse may fall out of the capture frame. E.g. if the capture pulse is near the right of the frame, a positive offset would push part of the pulse beyond the right limit of the frame. Suggest one technique that can be applied to the augmented data to filter off data that does not have a complete pulse.

one way is to derive avg energy Ex, compute energy of augmented waveform, if its similar to Ex then waveform should be captured and used

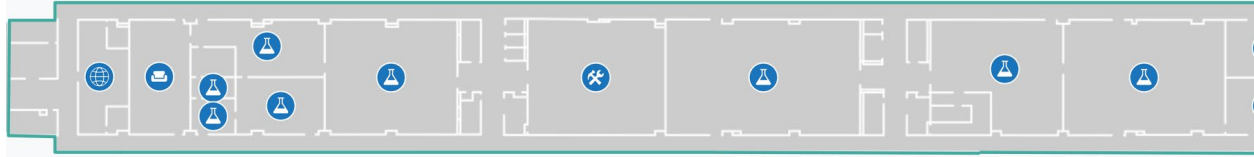
(iv) In actual deployment, the data capture may not be ideal, so the waveform may not be as illustrated in Figure Q1-1 but may have a horizontal offset to the left or right. One mitigating technique is to have sliding (and overlapping) windows within the capture frame to ensure essential data is fed into the NN model. Explain how this technique works. Suggest the duration of the capture frame, window size and time offset between windows. What are the trade-offs/factors to consider when deciding on these parameters?

same way as sliding window, smaller inference window across big capture window, smaller inference window = might miss waveform, bigger inference window = higher computation, smaller offset = more overlap in data, wasting space, big offset = might miss waveform.

(v) During NN model training, a typical way is to arrange the accelerometer and gyro data as a 1D tensor to feed into the NN model. Suggest other ways of arrangements and their advantages (if any).

There may be correlations btw adjacent samples of accele and gyro dataset wrt time, especially when gesture is performed across. 2D tensor can be evaluated as an image with say the CNN to extract the dependency features that will enable more accurate robot.

Q2: Indoor asset location tracking via Wifi RSSI Fingerprint



Source:NTUmaps

GPS based location tracking typically does not work well indoors as GPS signal are often weak or block when indoors. A more reliable means is via detection of strength of wifi signals from Wifi Access Points (AP) in the vicinity of the object. The list of strength of wifi signals from wifi AP detected by the asset tag (wifi receiver) attached to the object forms a pattern (signature) which is unique to the object's location, e.g. [AP1=-43, AP2=-66, AP3=-33] may indicate that the object is at location (x_k, y_k) . The continuously improving indoor location tracking performance of Apple and Google Maps is a result of training of data from GPS, Cellular and Wifi signals (plus other techniques). RSSI stands for Receiver Signal Strength Indicator, it is proportional to Strength of the RF Signals received.

(i) Common issues that developers may encounter during data collection are

Need measure in a longer time period (multiple days, diff time), to see which temporary AP will appear and disappear indicating temporary.

- Cannot differentiate between persistent and temporary Wifi signals. Temporary wifi signals may come from personal wireless routers, handphone hotspots etc.

no signal can be mitigated during training, by selectively shutting down certain percentage of AP's with dataset, similar to dropout concept

- No signals detected from certain Wifi APs.

What are the possible mitigating measures to counter these issues?

(ii) Typical data collected is a time series dataset of the RSSI of the surrounding APs. Simplest format of data arrangement (for training purpose) is $(x_k, y_k) \Rightarrow [AP1_T, AP2_T, \dots, APN_T, AP1_{T+1}, \dots, APN_{T+1}, \dots, APN_{T+M}]$, where $N=AP$ number, $T=time$, $M=time$ offset. Suggest other possible data formats and their pros/cons (if any).

The 2D tensor can be form by arranging data at each time instance as one column, this allows correlation information btw time instances to be included in training and inference in CNN model

(iii) Following from Q2(i)b above, if AP 1 is located very far away and its signal is too weak to be detected at location (x_k, y_k) , what RSSI value do we need to initialize $AP1_T$ to?

(iv) If Bluetooth-based Outpost is used instead of Wifi AP, and you have control over whether to design the asset tag (Bluetooth) to be a transmitter or receiver, which option would you choose and why? Discuss from perspective of power consumption. Asset tag is battery powered and Bluetooth Outpost is main powered.

iii) some unique number that can never be achieved from weak signals, as long as it is able to be differentiated from different locations. thus a value like 100 works

iv) polling for receiving RF data is very intensive, tags being battery powered should be a transmitter as transmitting takes a set amount of power can be controlled to determine how long the battery can last, however, if receiving RF constantly polling is very intensive thus it consumes more power.